ELECTRONICS GUIDE

A Comprehensive Guide to Varistor Selection & Application

From datasheet parameters to practical protection design

In this booklet we learn — in simple but precise language — what a varistor is, what it is made of and how it works; we read its V–I characteristic curve, interpret the key datasheet parameters, weigh the four decisive selection parameters with numerical examples, and practise, step by step, choosing a suitable varistor for protecting equipment.

Fundamentals & Physics | V–I Curve | MOV | Protection Circuit | 4 Selection Parameters | Recommended Models | Worked Example

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Abstract — The varistor is one of the most widely used protective components in power electronics: with its non-linear behaviour it absorbs the energy of transient overvoltages and saves the life of the equipment. This booklet is written for electrical and electronics students and for repair technicians; its only prerequisites are a basic familiarity with Ohm's law and with AC circuits.

SECTION 1

What Is a Varistor?

A varistor is a component whose name is built from the two words **Variable** and **Resistor**; it is also called a "voltage-dependent resistor", or **VDR**. Unlike an ordinary resistor, which has a fixed value, the electrical resistance of a varistor depends strongly on the voltage applied across it: at low voltages, within the normal operating range, the varistor is a very large resistance (in the megaohm range) and has practically no effect on the circuit; but the moment the voltage passes a defined threshold, its resistance suddenly collapses to a few milliohms and, like an escape path, it diverts the attacking current to ground.

The behaviour of a varistor can be likened to the safety valve of a steam boiler: as long as the pressure stays within the permitted range, the valve is closed and inert; but the moment the pressure passes the critical limit, the valve opens and releases the excess steam so that the boiler does not explode. In the same way, the varistor is "off" under normal conditions and sits silently beside the circuit, but at the instant a transient overvoltage occurs it goes into action, absorbs and drains the destructive energy, and after the transient surge is over it returns to its normal state. It is exactly this two-way, automatic behaviour that has made it the most widely used protection component against transient overvoltages (surges).

◆ Why do circuits need protection?

Transient overvoltages usually arise from two main sources: natural phenomena such as lightning, which — through electromagnetic induction on transmission lines — creates voltages in the range of several kilovolts; and switching operations in the power grid, i.e. the instant of switching inductive loads such as motors, transformers and contactors which, because of the energy stored in inductors, produce short but intense transient overvoltages. If these transients reach a circuit unchecked, they can destroy sensitive semiconductors such as microcontrollers, drivers and power modules in a fraction of a microsecond; and even when a surge is not lethal, its constant repetition causes gradual erosion and shortens the life of the parts.

◆ Circuit symbol

In circuit schematics, as you can see in the figure below, the varistor is drawn as a resistor with a diagonal line through it; in the IEC standard the letter **U** is placed beside the diagonal to distinguish the voltage-dependent resistor from thermistors and ordinary resistors. In industrial designs the part is labelled **MOV1** or **RV**.

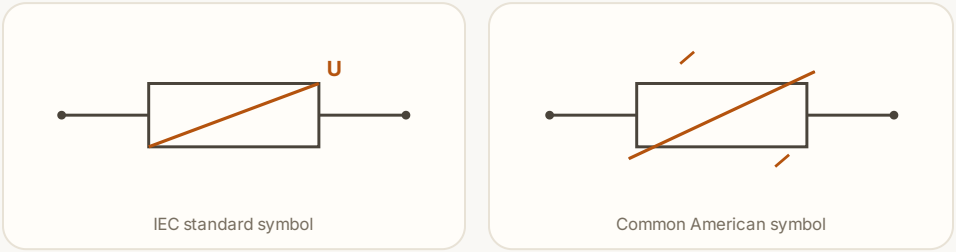


Figure 1 — Varistor circuit symbols in the two standards; the diagonal line is the mark of the resistance's dependence on voltage.

~MΩ	~mΩ	<25ns	AC/DC
Resistance at normal voltage	Resistance during overvoltage	Approximate response time	Bidirectional application

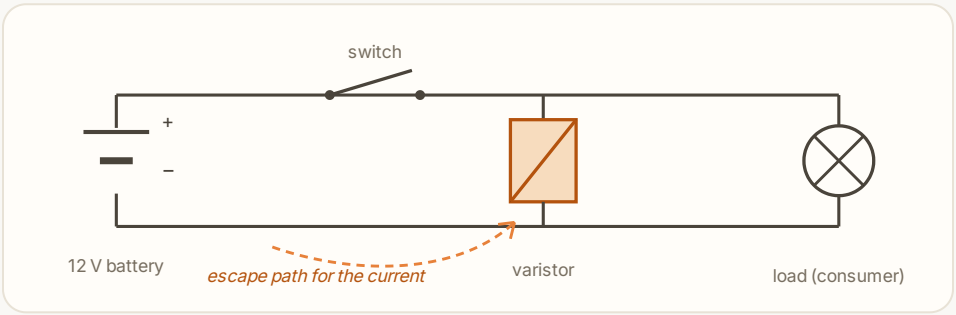


Figure 2 — The place of the varistor in a circuit: always in parallel with the load; it is inert in normal operation and becomes the escape path for the current during an overvoltage.

END OF SECTION 1

SECTION 2

Structure & Operating Physics

The most common type of varistor, which covers practically all industrial and domestic applications, is the **metal oxide varistor** or **MOV** (Metal Oxide Varistor). To build it, zinc oxide (ZnO) powder is mixed with additives such as bismuth, cobalt, manganese and nickel oxides, then pressed and sintered at about 1200 °C. The result is a ceramic disc whose two faces are coated with a metal electrode and to which radial leads are soldered.

The secret of the MOV's non-linear behaviour lies in its grain structure. Under the microscope, the body of the varistor is made of millions of conducting zinc oxide grains, each of which is surrounded by a thin oxide layer rich in bismuth — the **grain boundary**. These boundaries behave like reverse-biased PN junctions; that is, each boundary forms a **potential barrier** with a breakdown voltage of about 3.2 to 3.5 V. If the current path crosses dozens of grains, the series connection of these same barriers builds the **varistor voltage** (Varistor Voltage – V1mA); the manufacturer therefore sets the varistor voltage with the thickness of the disc (the number of grains in the path) and the energy absorption capacity with its diameter (the number of parallel paths).

When the applied voltage is below the threshold, the electrons cannot pass the potential barriers of the boundaries and only a leakage current in the microampere range remains. But when the voltage passes the threshold, the electric field across the boundaries grows strong enough for the barriers to collapse; the current increases steeply through tunnelling and avalanche-drift mechanisms and the apparent resistance of the part falls by several orders of magnitude. The process is completely reversible and, after the transient surge ends, the varistor returns to its high-resistance state — provided that the energy passing through it has not exceeded its permitted capacity.

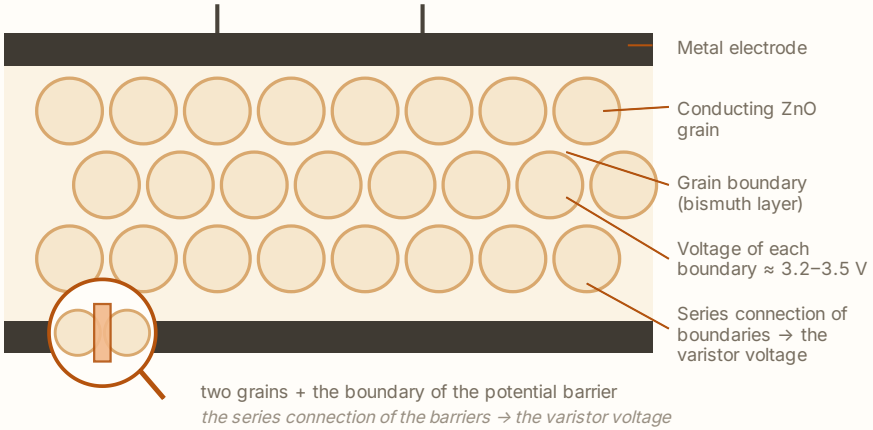


Figure 3 — Internal cross-section of a metal oxide varistor; the conducting ZnO grains and the insulating grain boundaries between them.

Key point — The varistor is a completely bidirectional element: its grain structure and potential barriers have no polarity, so it works equally in both half-cycles of an AC wave — and this very feature makes it an ideal protector for AC circuits.

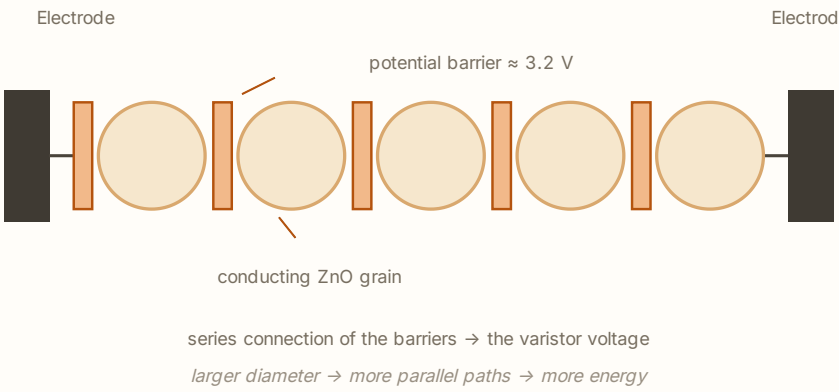


Figure 4 — The equivalent circuit of the varistor: the series connection of the potential barriers (reverse-biased junctions) builds the varistor voltage.

END OF SECTION 2

SECTION 3

The V–I Characteristic Curve & Operating Regions

The most important tool for understanding the behaviour of a varistor is its voltage–current characteristic curve. Because the resistance of the varistor changes by several orders of magnitude, this curve is drawn on logarithmic axes so that both the region of microampere currents and that of kiloampere currents are seen in one frame. The curve is symmetric about the origin; that is, the varistor behaves identically at positive and negative voltages. In general, three distinct operating regions exist:

◆ 1) Leakage region (pre-breakdown)

At voltages below the threshold, the varistor behaves like a very large resistance (in the megaohm range) and only a leakage current of a few microamperes passes through it. In this region the curve is almost linear and the varistor looks practically “inert”. Note that if you feed a varistor permanently with a voltage close to the threshold, its leakage current and heating increase and its life is shortened; for this same reason, a safety margin between the working voltage of the network and the varistor voltage is always necessary.

◆ 2) Breakdown (non-linear) region — the main workplace

Once the voltage passes the threshold, the current grows with the power relation $I \propto V^{\alpha}$, in which the non-linearity exponent α for zinc oxide varistors takes a very large value (about 30 to 50). That is, a rise of a few tens of percent in the voltage multiplies the current thousands of times. In this very region the varistor absorbs the energy of the transient surge and “clamps” the voltage across the load at a defined ceiling, and then, after the surge has drained away, it returns to the leakage state.

◆ 3) Upturn region

If the current grows beyond the capacity of the non-linear region itself, the intrinsic and series resistance of the varistor body (the ceramic material and the electrodes) takes over and the curve becomes linear and sloped again. In this region the varistor no longer clamps the voltage and the incoming energy turns mostly into heat; continuing to work in this region without limitation leads to cracking, burning and even explosion of the body. For this same reason, a current-limiting element (a fuse or a resistor) must always be placed before the varistor.

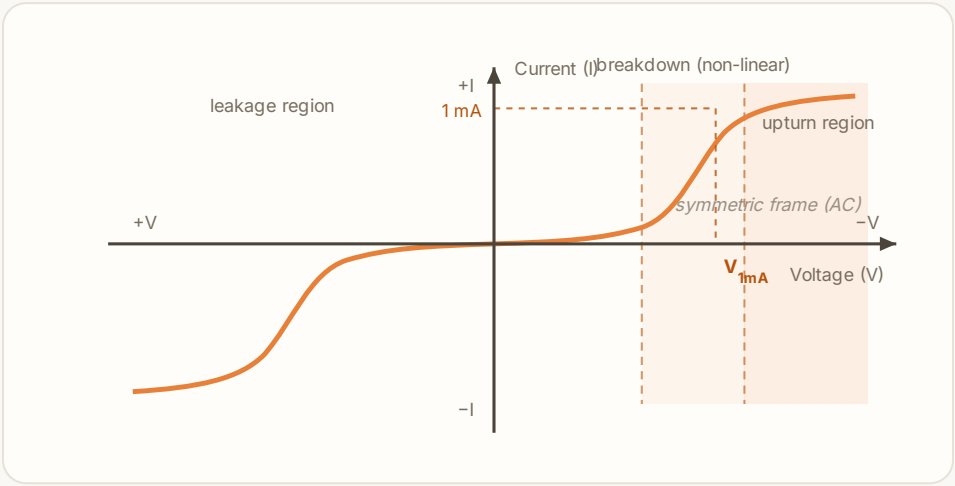


Figure 5 — The V–I characteristic curve of the varistor (logarithmic axes); the point V_{1mA} is the varistor voltage at a current of 1 milliampere.

Section summary — the leakage region = “off”; the breakdown region = “the workplace and the absorption of energy”; the upturn region = “danger and high temperature”. Correct selection means making sure that the real transient surges always drive the varistor only to the end of the non-linear region — and no further.

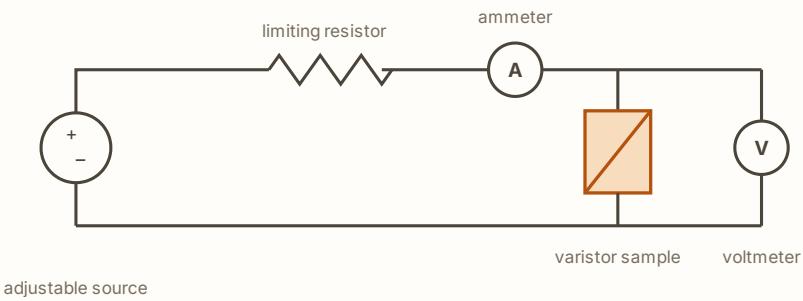


Figure 6 — The circuit for measuring the varistor voltage: the voltage at which the current reaches 1 milliampere is called V_{1mA}.

END OF SECTION 3

SECTION 4

Key Datasheet Parameters

To select a varistor correctly you must know the language of the manufacturer's **datasheet**. The table below lists the most important parameters together with their definitions and the sample values of a 14 mm disc with a 275 V rating (for a 230 V network). Note that the exact values differ slightly from one manufacturer to another, and you must always refer to the datasheet of the very product at hand.

Two points in reading this table are vital. First, the "maximum continuous operating voltage" is **not** the same thing as the "varistor voltage": the former is the voltage that may be applied all the time, while the latter is only a defining point on the curve. Second, the "clamping voltage" is always several hundred volts above the varistor voltage; the protected circuit must therefore be able to tolerate this very residual voltage at the moment of the surge — varistor protection means "reducing" the overvoltage to a tolerable level, not removing it completely.

Parameter	Symbol	Definition and sample value
Non-linearity exponent	α	The criterion of the sharpness of the non-linear behaviour in the breakdown region; about 30 to 50 for zinc oxide (the higher, the better)

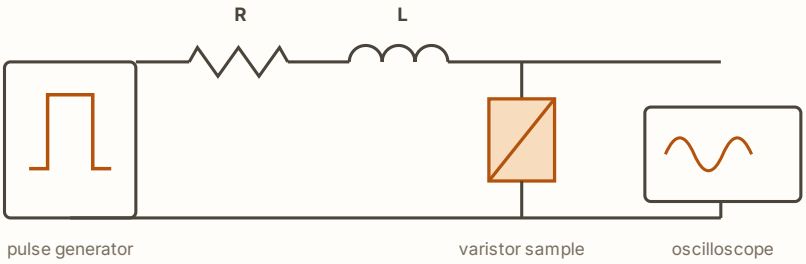


Figure 7 — The test arrangement of the standard 8/20 µs surge; the current capacity and the clamping voltage of every varistor are obtained with this same circuit in the datasheet.

END OF SECTION 4

SECTION 5

Varistor Types & Comparison with Other Protectors

Over time, varistors have taken many forms. The first generation was the **silicon carbide (SiC)** varistor, which was used in the grid arresters of the middle decades of the twentieth century; its lower non-linearity exponent and its need for a series resistor have made it obsolete today. The second generation, which dominates the market, is the same **metal oxide varistor (MOV)**, built as disc and block parts. The third generation is the **multilayer chip varistor (MLV)**, made as SMD parts and used for protecting fast data lines such as USB, HDMI and antennas.

In protection design the varistor is not the only option and is usually compared with two other parts: the **TVS diode**, a fast and precise but low-capacity protector, and the **gas discharge tube (GDT)**, which has a very high discharge capacity but a lower speed and lower precision. The table below is a practical comparison of these three parts:

Criterion	Varistor (MOV)	TVS diode	Gas discharge tube (GDT)
Response speed	Very fast (<25 ns)	Extremely fast (~ps–ns)	Slower (hundreds of ns and more)
Current / energy capacity	High (kiloamperes)	Low (watts to kilowatts of surge)	Very high
Clamping voltage precision	Medium	Precise, with low deviation	Low, and dependent on the rate of rise of the surge
Normal leakage current	Microampere	Very low	Practically zero
Parallel capacitance	A few hundred pF	Low (a few tens of pF)	Practically zero
Sample application	Mains input, power boards	Sensitive data and logic lines	Large arresters, telecom lines

Practical rule — in industrial equipment a combination is usually used: an MOV at the mains input to absorb the bulk of the energy, then a TVS near the sensitive part for a more precise clamp, and in very heavy cases a GDT ahead of the MOV to drain the main energy. The three are complements of one another, not substitutes.

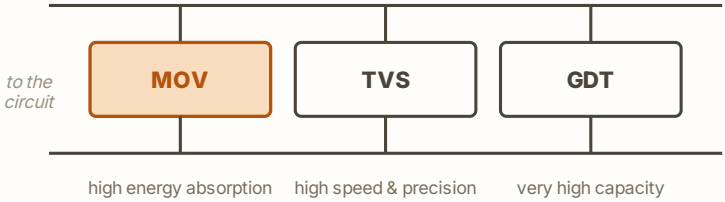


Figure 8 — A practical comparison of the three overvoltage protectors in a circuit; all three are connected in parallel with the source, and their difference lies in speed, precision and energy capacity.

END OF SECTION 5

SECTION 6

Applications & Protection Circuits

The varistor is always connected **in parallel** with the load, never in series, because its job is to create a low-resistance escape path for the surge current. In the figure below you see the simplest and most common protection structure: a fuse in series in the line and a varistor between line and neutral. In the normal state the current flows from the fuse to the load and the varistor draws almost no current. At the moment of the transient surge, the varistor conducts, draws the resulting current and holds the voltage across the load at the clamping ceiling; the large current of this very path blows the fuse and separates the load from the network.

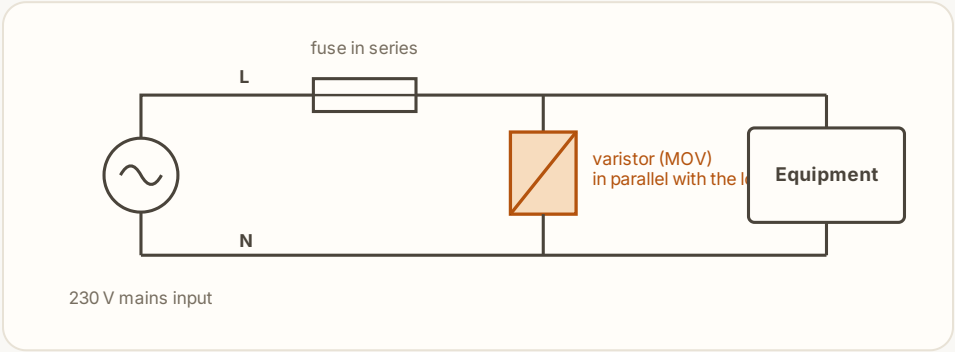


Figure 9 — The basic protection structure: a fuse in series in the line and a varistor in parallel with the load, between line and neutral.

The role of the fuse can never be removed, because the usual failure mode of the varistor after a very heavy transient surge is a **permanent short circuit**. Without a fuse, a burnt varistor keeps drawing the network current, heats up and can lead to fire; for this same reason, standard products combine the varistor with a **thermal fuse** in a single package, so that at the end of the part's life the circuit is disconnected safely.

Where you will see a varistor

- **Switching power supplies**
  On the mains input, before the bridge rectifier; the first line of defence of the whole power supply.

- **Protected sockets and switches**
  In household overvoltage and short-circuit protectors, the heart of the device is a single MOV.
- **Industrial boards and PLCs**
  Protecting the command inputs and the relays against the inductive switching transients.
- **Solar and telecom equipment**
  Inverters and outdoor equipment that are exposed to lightning induction and to network transients.

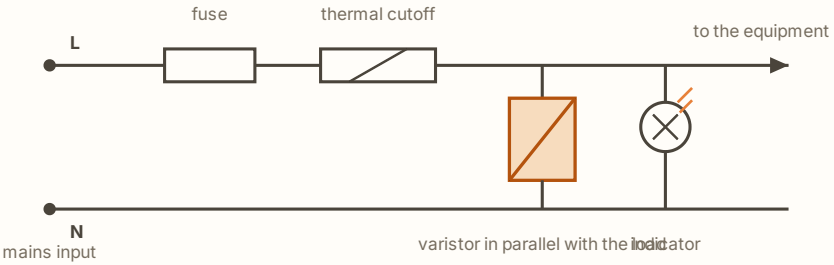


Figure 10 — The safe connection of the varistor in real equipment: a series fuse and a thermal cutoff before the varistor, and an indicator in parallel with it.

END OF SECTION 6

SECTION 7

The Four Key Parameters for Selecting a Varistor

When selecting a varistor, looking at the network voltage alone is not enough; four parameters must be examined at the same time and matched to the very circuit at hand: the continuous operating voltage (MCOV), the clamping voltage (V_C), the maximum surge current (I_{TP}/I_{peak}) and the absorbable energy (W/J). In this section we review all four parameters with numerical examples, based on the datasheet of the widely used part **Bourns 14D471K** (a 14 mm disc for a 220 V network).

◆ 1) Continuous operating voltage — MCOV

The maximum voltage that can be applied to the terminals of the varistor continuously and uninterruptedly, without it entering conduction, heating up or having its life shortened. The datasheet quotes two separate numbers — one for AC and one for DC. According to the Bourns datasheet for the 14D471K: **300 VAC / 385 VDC**.

Why two different numbers? In AC, the peak voltage (about 424 V for 300 VAC) is applied only for an instant in each cycle; but in DC the same voltage is present continuously and creates continuous thermal leakage. For this reason the DC limit is defined higher than the AC limit — yet both must stay, with a safety margin, below the varistor voltage (470 V).

Why is it critical? — If the circuit voltage rises above the MCOV, the varistor leaks permanently ← it heats itself up ← its resistance falls even lower ← more heat ← **thermal runaway** and, in the worst case, fire. It is exactly for this reason that using the MOV together with a thermal cutoff is recommended.

◆ 2) Clamping voltage — V_C (Clamping Voltage)

The clamping voltage is the maximum voltage that the varistor — and therefore the circuit behind it — shows at the moment of the surge. The common misconception is that when the varistor "turns on" it zeroes the voltage like a switch; it does not! The varistor is a non-linear resistor, not a wire, and while the surge current flows it has a voltage drop of its own. V_C is exactly the "ceiling" that the circuit voltage reaches at the worst instant of the surge, and the circuit behind the varistor sees precisely this same voltage.

Key note: V_C is not a fixed number! In the datasheet, the clamping voltage is always quoted together with the test current, because the higher the surge current, the higher the voltage across the varistor rises as well. The expression **775 V @ 50 A** means: if the standard 8×20 µs surge wave with a peak of 50 amperes passes through the 14D471K, the voltage across it becomes at most 775 volts.

Selection example for a 220 V network

- The 220 V network with a ±10% tolerance ← maximum continuous voltage **242 V**.
- **Rule:** minimum MCOV ≥ 1.1 × maximum circuit voltage ← at least ≈**270 VAC** is needed.
- **Options:** 14D431K (275 VAC, clamping 710 V) or 14D471K (300 VAC, clamping 775 V) ✔
- The same rule for DC systems too: a 24 V battery (charged up to ≈29 V) ← a varistor with MCOV(DC) ≥ 31 V, such as **MOV-14D390K**.

MCOV trade-off note — do not raise the MCOV without a reason! A varistor with a higher operating voltage also has a higher clamping voltage and gives weaker protection — exactly the story of the 14D561K that we will see shortly.

Step-by-step example: a 220 V power supply whose parts tolerate 900 V

- **Normal state:** the peak of the 220 V RMS network is about 311 V; far below 470 V ← the varistor is practically an open circuit and has only a slight leakage; nothing happens. ✔
- **The surge enters:** the instantaneous voltage rises; as soon as it passes ≈470 V, the varistor conducts rapidly and drains the surge current.
- **The peak of the surge:** with a peak current of 50 amperes, the voltage across the varistor clamps at ≈775 V.
- **Final verdict:** the parts of the circuit see ≈775 V instead of the thousands of volts of the surge, and since 775 < 900, they survive. ✔

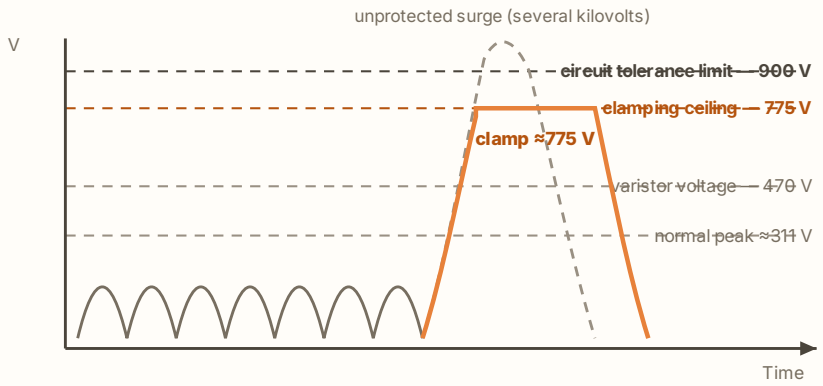


Figure 11 — The clamping voltage: the varistor limits a surge of several kilovolts to a ceiling of about 775 volts; since 775 is below the tolerance of the circuit (900 V), the parts survive.

Counterexample — when the clamping voltage is ignored

- Suppose that instead of the 14D471K a **14D561K** has been used (varistor voltage 560 V and operating voltage 355 VAC — as far as the first parameter is concerned it has no problem with the 220 V mains). But the clamping voltage of this part at the same 50-ampere current is about **925 V**; 925 > 900 ← in the very surge that was life-saving in the previous example, the MOSFET now burns!
- That is, the varistor stays "healthy" but fails to do the job we asked of it. So the clamping voltage must be checked against the tolerance of the circuit, not only against the operating voltage.

Two practical notes

- **Bigger surge current ← higher clamp:** if the probable surge of your circuit is not 50 A but several hundred amperes, V_C rises above 775 V and may break the tolerance threshold of the circuit; for this same reason, parameters 2 and 3 must be examined together.
- **Bigger diameter ← lower clamp at the same current:** if with a 14D you cannot reach the clamping voltage you want, a larger size (such as 20D471K) at the same surge current has a lower current density and delivers a lower clamp — plus a greater absorbed energy.

Summary in one sentence — the clamping voltage is the "voltage ceiling" that your circuit will have to tolerate at the worst moment of the surge; correct selection therefore means: V_C (at the real surge current of the circuit) below the tolerance of the sensitive parts.

◆ 3) Maximum surge current — I_{TP} / I_{peak}

The maximum peak surge current with the standard 8×20 µs wave shape that the varistor withstands without cracking or being destroyed; for the 14D471K it is **4500 A** according to the datasheet. Why is it critical? Because a surge current beyond the limit cracks the ceramic ZnO disc from the inside, and the result is arcing, smoke and the explosion of the part. The varistor is the "hero" of the circuit, but it has a defined capacity!

Important note — the number of surges: the figure 4500 A belongs to a few (1–2) surges. If the surges are repeated (say a hundred surges over the life of the equipment), the permitted current becomes many times lower (the derating curve in the datasheet). For a long life, always apply a safety factor.

Step-by-step numerical example

- A surge source with an open-circuit voltage of 6 kV and an internal impedance of 2 Ω enters the circuit.
- Probable surge current: **I = 6000 ÷ 2 = 3000 A**.
- Comparison: the 14D471K with I_{TP} = 4500 A ✔ passes; but the same varistor in the smaller size (the 10D series with a capacity of ≈2500 A) ✖ would burn.
- **Rule:** I_{TP} must be greater than the maximum probable surge current at the point of installation — and this figure depends on the place of installation, the cable length and the class of protection.

◆ 4) Absorbable energy — W / J

The total energy that the varistor absorbs during the surge and dissipates as heat; that is, the time integral of the instantaneous voltage across the part multiplied by the flowing current ($W = \int v \cdot i \, dt$). For the 14D471K, the Bourns datasheet declares **125 J** (in the long 10/1000 µs wave).

Key note — the dependence on the wave shape: the permitted energy is not an absolute number and changes with the shape of the surge wave. Long waves (the switching of motors, the blowing of a fuse, the interruption of an inductor) carry far more energy despite their lower current, and are more dangerous. The rule: the energy of the probable surge of the circuit (with a similar wave shape) must be less than the permitted energy of the varistor; otherwise the varistor overheats and is destroyed.

◆ The four parameters must be confirmed "together"

None of these four conditions is sufficient on its own; a part is selectable only when all four rows of the table below hold for it at the same time:

Parameter	Symbol in the datasheet	14D471K value	Acceptance rule
Continuous operating voltage	**MCOV**	**300 VAC / 385 VDC**	MCOV ≥ 1.1 × maximum circuit voltage
Clamping voltage	**V_C**	**775 V @ 50 A**	V_C < tolerance limit of the circuit parts
Maximum surge current	**I_{TP} / I_{peak}**	**4500 A (8×20 µs)**	I_{TP} > probable surge current at the point of installation
Absorbable energy	**W**	**125 J (10/1000 µs)**	Rated energy > surge energy of the circuit

A single example — the same 220 V circuit with parts rated 900 V and a probable surge of 3 kA: the surge energy is obtained from the approximate relation $E \approx V_C \times I \times \tau$; that is, 775 V × 3000 A × 20 µs ≈ 47 J. Now put all four conditions side by side:

Condition	Circuit value	14D471K capacity	Status
Continuous operating voltage	**maximum 242 V (from 220 V ±10%)**	**300 VAC**	**✔ Confirmed**
Clamping voltage	**775 V < 900 V**	**775 V @ 50 A**	**✔ Confirmed**
Surge current	**3000 A**	**4500 A**	**✔ Confirmed**
Absorbed energy	**≈47 J**	**125 J**	**✔ Confirmed**

Result — all four conditions hold ← the **14D471K** is approved for this circuit. If even one of the rows had failed (like the 14D561K, whose clamping row failed), we would have had to choose another size or model. ✔ All four parameters must be matched with the circuit — not just one or two of them.

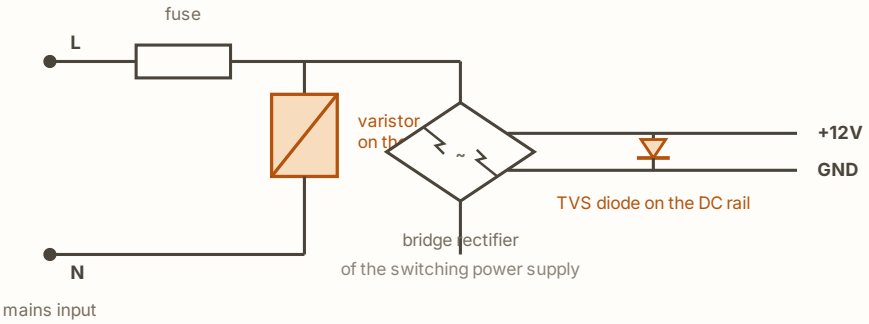


Figure 12 — The input protection of a switching power supply: the varistor on the mains input and the TVS diode on the DC rail.

END OF SECTION 7

SECTION 8

Selecting a Varistor: a Worked Example for Mains Power

We want to select a varistor for a piece of equipment connected to a single-phase 230 V AC network. The golden rule is that the "maximum continuous operating voltage of the varistor" must be at least 10 to 25 percent higher than the maximum normal voltage of the network, so that the leakage current and the heat of the part stay within the safe range. Follow the steps:

1. **Determine the normal voltage of the network.** A 230 V RMS network with a permitted fluctuation of 10% rises up to 253 V; the peak of the normal voltage is also $\approx\sqrt{2} \times 230 = 325$ V.

2. **Choose the operating voltage rating.** MCOV $\geq 1.1 \times 230 \approx 253$ V ← the nearest standard rating is the **275 V AC** varistor.

3. **Check the varistor voltage.** The varistor voltage $V_{1mA} \approx 430$ V is, with a safe distance, above the network peak (325 V); so in normal operation the varistor remains in the leakage region.

4. **Estimate the transient surge and the capacity.** For a usual industrial environment, the expected surge current is a few kiloamperes (8/20 µs); the 14 mm disc with $I_{max} \approx 4.5$ kA is selected, and for high-risk places a 20 mm disc or two discs in parallel.

5. **Complete the protection circuit.** A suitable series fuse before the varistor, enough air distance and thermal path on the board, and where needed, a clamping voltage of ≈710 V is checked against the tolerance of the load.

Selection result — a 14 mm disc rated **275 V AC** (in the style of the S14K275); varistor voltage $V_{1mA} \approx 430$ V, clamping ≈710 V @ 50 A, surge current ≈4.5 kA, together with a series fuse. For 110–127 V networks the 150 V or 175 V rating is used, and for DC circuits the DC voltage rating of the datasheet.

◆ Common design mistakes

1. Selecting a varistor with a rating close to the network voltage (e.g. 230 V for a 230 V network) ← leakage, permanent heat and premature failure.
2. Removing the series fuse ← the risk of a permanent short circuit after a heavy transient surge, and fire.
3. Ignoring ageing — after repeated surges the varistor voltage falls and the part must be inspected and replaced.
4. Relying exclusively on the MOV against direct lightning ← a high-class arrester and a GDT are needed as well.
5. Using a varistor on fast data lines without paying attention to its capacitance ← attenuation of the pulse and errors of the signal.

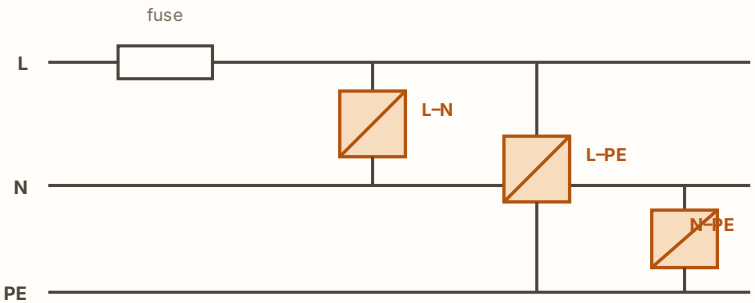


Figure 13 — The standard protection arrangement in a single-phase panel: three varistors between L–N, L–PE and N–PE, together with the series fuse.

END OF SECTION 8

SECTION 9

Recommended Varistor Models for Common Voltages

In this section, several widely used models that are available on the market are introduced for the two most frequent applications; all the figures are taken directly from the manufacturers' datasheets (the Bourns MOV-xxD series and the AVX TranGuard). The logic of the selection is the same four parameters of Section 7: a continuous operating voltage at least 10% above the maximum circuit voltage, a clamping voltage below the tolerance of the parts, and a surge current and permitted energy greater than the probable hazards of the installation point.

◆ Low-voltage DC circuits — 5, 12 and 24 volts

Circuit voltage	Recommended model	MCOV (DC)	V_{1mA}	Clamping V_C	I_{max} / energy
5 V (logic and microcontroller)	AVX VC080505A150 — MLV 0805	**5.6 V**	**8.5 V**	**18 V @ 1 A**	**40 A / 0.1 J**
12 V regulated (power supply)	MOV-14D180K (smaller: 07D180K)	**14 V (11 VAC)**	**18 V**	**36 V @ 10 A**	**1 kA / 4 J**
12 V automotive / battery (charged to 14.4)	MOV-14D220K	**18 V**	**22 V**	**43 V @ 10 A**	**1 kA / 5 J**
24 V battery / industrial (charged to 28.8)	MOV-14D390K (smaller: 07D390K)	**31 V (25 VAC)**	**39 V**	**77 V @ 10 A**	**1 kA / 8.6 J**

Why is there no "disc" for 5 volts? — MOV discs are not made below a varistor voltage of 18 V; for 5 V lines a multilayer chip varistor (MLV) must be used. If you want a larger voltage margin, the VC080509A200 (clamping 22 V) is a better choice; and if you have a sensitive data line, note that the parallel capacitance of these parts (≈1100 pF) attenuates fast pulses — on such a line the TVS diode is more precise.

The three-phase 380 V network

Connection method	Model (on each phase)	MCOV	V_{1mA}	Clamping V_C	I_{max} / energy
3× phase–neutral (220 V)	MOV-14D431K	**275 VAC / 350 VDC**	**430 V**	**710 V @ 50 A**	**4.5 kA / 115 J**
3× phase–phase (380 V)	MOV-14D681K	**420 VAC / 560 VDC**	**680 V**	**1120 V @ 50 A**	**4.5 kA / 130 J**
High-risk place (lightning / heavy switching)	3× MOV-20D681K	**420 VAC / 560 VDC**	**680 V**	**1120 V @ 100 A**	**6.5 kA / 220 J**

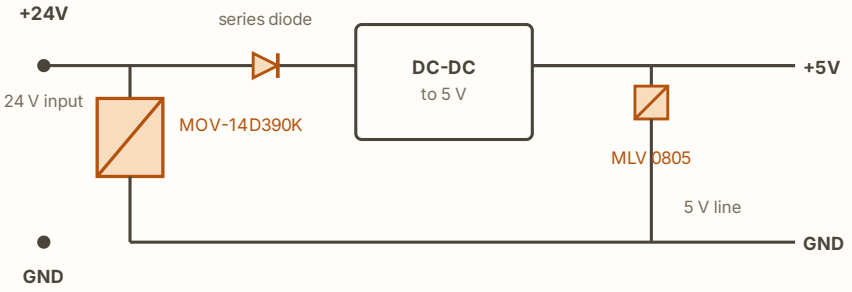


Figure 14 — Protection of a DC line with the recommended models of Section 9: a varistor on the 24 V input and an MLV on the 5 V line near the sensitive load.

- In the phase–phase connection, the 621K rating (385 VAC) is permitted only for a network with a stable voltage; with a +10% tolerance (maximum 418 V) the 681K (420 VAC) must be used.
- The phase–neutral connection is the same 275 V rating as Section 8; since each phase clamps independently, it provides more effective protection for the single-phase loads inside the panel.
- For more energy, use a 20 mm disc or two 14 mm discs in parallel (each with its own independent series fuse).
- The equivalents of these ratings also exist with other manufacturers (Littelfuse, TDK SIOV, Thinki); before buying, check the MCOV, V_C and I_{max} columns of the datasheet of the very product.

END OF SECTION 9

SECTION 10

Summary & Safety Notes

The five points to take away from this booklet

1. A varistor is a voltage-dependent resistor: it is off at the normal voltage and conducting during the overvoltage.
2. The heart of the MOV is the ZnO grains and the bismuth grain boundaries; the series connection of the potential barriers builds the varistor voltage (Varistor Voltage – V_{1mA}).
3. The V–I curve has three regions: leakage, non-linear breakdown (the workplace) and upturn (the danger region).
4. The key parameters: MCOV, the varistor voltage V_{1mA}, the clamping voltage V_C, and the surge current and energy.
5. Always connected in parallel with the load, together with a series fuse; in heavy applications, the MOV + TVS + GDT combination.

Safety — the varistor becomes hot while draining heavy transient surges and may crack or explode; when replacing it, let it cool down, use a part of the same rating and the same size, and never choose a voltage rating lower than the calculated one. Before working on a circuit, disconnect the power and discharge the capacitors.

Key vocabulary of this booklet

Term	What does it mean?
VDR	Voltage-dependent resistor; the other name of the varistor
MOV	Metal oxide varistor; the most common type of varistor, with a zinc oxide core
8/20 µs	The standard surge wave shape for measuring the discharge current of the varistor
Aging	Ageing; the gradual drop of the varistor voltage after repeated surges
V_C / Clamping Voltage	The clamping voltage; the ceiling of the voltage across the varistor while the surge current flows
MCOV	Maximum continuous operating voltage; quoted in the datasheet separately for AC and DC
Potential Barrier	Potential barrier; the ≈3.2-volt barrier at every grain boundary, whose series connection builds the varistor voltage
MLV	Multilayer SMD varistor for low-voltage and data lines

- The varistor is a voltage-dependent resistor: off at the normal voltage, conducting at the overvoltage.
- The heart of the MOV is the ZnO grains and the bismuth boundaries; the series connection of the potential barriers builds the varistor voltage (V_{1mA}).
- The V–I curve has three regions: leakage, non-linear breakdown (the workplace) and upturn (danger).
- The key parameters: MCOV, the varistor voltage V_{1mA}, the clamping voltage V_C, and the surge current and energy.
- Always connected in parallel with the load, with a series fuse; in heavy applications, the MOV + TVS + GDT combination.

With this knowledge you can read the datasheet of any varistor, recognise its role in the schematics, and design a reliable protection for any piece of equipment; proper protection is always cheaper than the repair.

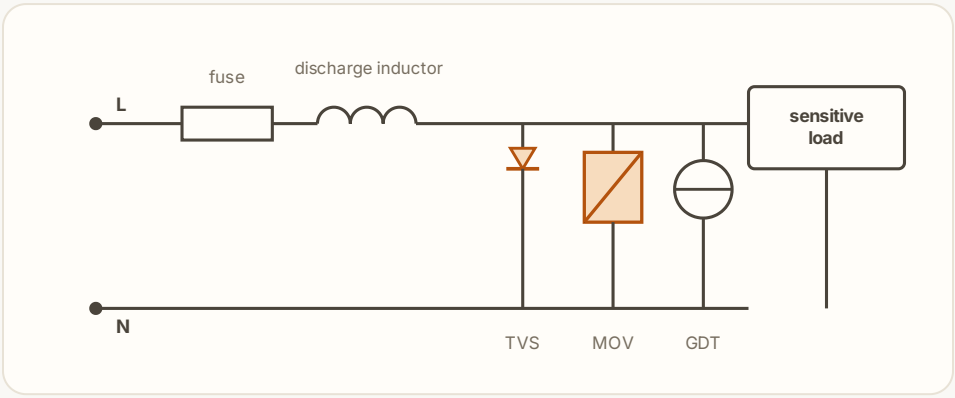


Figure 15 — Summary: the complete protection chain; every link is designed for a specific threat, and the series fuse is never removed.

END OF SECTION 10

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About the contents of this booklet and guide, we would of course be glad to hear your viewpoints and suggestions. Please also share your corrections, criticisms and questions on this subject with us.

With thanks, **Hassan Memari**

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